

WASP-43b

R-0465

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_240120 · created 2026-07-19T23:05:48+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460329.9912 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.013
Transit depth (Rp/Rs)^2	3.9 +/- 0.52 [%]
Orbital Inclination (inc)	85.25 +/- 1.33
Airmass coefficient 1 (a1)	0.97 +/- 0.029
Airmass coefficient 2 (a2)	0.016 +/- 0.011
Scatter in the residuals of the lightcurve fit is	2.93 %
Best Comparison Star	#3 - [179, 98]
Optimal Aperture	4.98
Optimal Annulus	15.78
Transit Duration (day)	0.0602 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.197 $\pm$ 0.013 vs 0.1594 (deviation 2.43 σ)
Duration fit vs published	0.0602 d vs 0.0512 d (deviation 3.42 σ)
Mid-time O-C	-2.1 min
Depth SNR	12.7
Residual scatter	2.93 %

Light curve

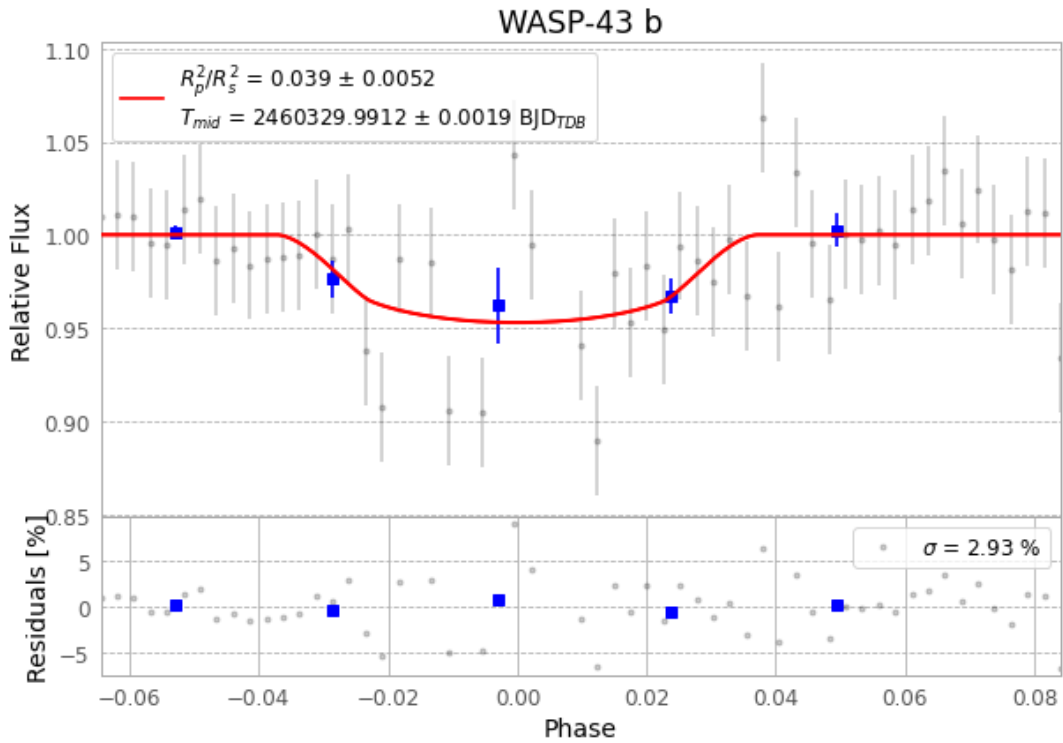


Figure 1 — FinalLightCurve_WASP-43 b_2024-01-20.png (SHA-256 cc6451753ed8cb9c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [477, 219], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 be1cad8e2a694db1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

59 FITS frames (39 MB), WASP-43240120102415.FITS → WASP-43240120131816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240120.log (b58d89be93c3...), FinalLightCurve_WASP-43 b_2024-01-20.png (cc6451753ed8...), FinalLightCurve_WASP-43 b_2024-01-20.csv (5137aa9e381d...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 23:05Z.